

Operational Insights into Alkaline Electrolyzers: Correlating Electrical Characteristics with Hydrogen Generation Efficiency

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July 21, 2025

Acknowledgements

I am writing to express my sincere gratitude to Mr. Selva and Mr. Inbaraj at Research Support India for their guidance and instruction during my internship. Their explanations of alkaline electrolyzers, electrode behavior, and key electrochemical processes were essential to my understanding and analysis.

I also acknowledge the use of Grammarly in assisting with proofreading and grammatical corrections throughout this paper.

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1. Abstract

Recently, I completed a field internship with Research Support India (RSI), an alkaline electrolyzer manufacturer based in Hosur, India. RSI serves major clients, including Reliance Industries and several international companies. During my time there, I gained hands-on experience with the principles and operation of alkaline electrolyzers. I was introduced to core electrochemical concepts, including the role of potassium hydroxide (KOH) as an electrolyte, membrane differences, and how the anode and cathode function to separate water into Hydrogen and oxygen gases. Through observation and guided instruction, I analyzed real-time operational data provided by the site's engineers, highlighting the relationships between voltage, current, power, and gas output. My analysis revealed patterns, such as the linear relationship between voltage and current, which helped me assess the system's performance. This experience deepened my understanding of clean energy technology and demonstrated the practical challenges and potential of Hydrogen as a scalable renewable energy solution.

2. Introduction

Recently, Hydrogen has been increasingly used as a clean energy resource as it does not produce carbon emissions. Alkaline water electrolysis, a well-established and straightforward method for hydrogen production, uses an alkaline solution, typically potassium hydroxide (KOH), to split water into Hydrogen and oxygen gases. The effectiveness and safety of this process rely heavily on controlling electrochemical reactions and electrical parameters such as voltage, current, and power. Despite being a mature technology, alkaline electrolyzers still require detailed study to better understand how these electrical factors impact hydrogen production efficiency and the overall stability of the system during operation. With guidance from engineers at RSI and my

analysis of real-time data, I explored how electrical parameters affect hydrogen production during hands-on work at a hydrogen energy facility.

The central research question addressed is: *How do electrical parameters such as voltage, current, and power correlate with hydrogen production efficiency and system stability in an alkaline water electrolyzer?* The key takeaways from this study aim to inform improved operational strategies and contribute to the advancement of hydrogen energy technologies.

Note: All scientific concepts and operational insights presented in this paper are based on technical instruction and direct observation during a field internship at Research Support India (RSI). I did not use any other external literary sources.

3. Color Classification of Hydrogen Production Methods

During my internship, I kept coming across terms like “green hydrogen” and “blue hydrogen,” so I researched how the industry actually classifies production methods. Engineers at RSI explained that the colors don’t literally describe hydrogen but instead indicate how clean the production process is. For example, grey hydrogen usually comes from natural gas and has very high emissions, while blue hydrogen is also natural gas-based but uses carbon capture to reduce pollution. The cleanest version I learned about was green hydrogen, which relies entirely on renewable power to split water molecules. However, they also noted that it’s the most expensive option. Other categories, like pink hydrogen (nuclear-powered) or yellow hydrogen (grid-powered), depend on the source of electricity. Although this color-coding is a simplification, it was a helpful framework for me to compare where alkaline electrolyzers fit in the bigger energy picture.

Color	Source and Method	Emissions	Notes
Grey Hydrogen	From natural gas via methane(CH_4)	High Carbon Emissions	The most widely used form of Hydrogen Production
Blue Hydrogen	From natural gas production, Carbon is captured and sent underground	Moderate Carbon Emissions	Cleaner than grey, but still fossil-derived
Green Hydrogen	Electrolysis powered by renewable energy sources (Wind, Solar)	Zero Carbon Emissions	Most sustainable, but also the most expensive
Black/Brown	Made from Coal or Oil via gasification	Highest Carbon Emissions	The most polluting form; even banned in some countries
Pink Hydrogen	Electrolysis powered by nuclear energy sources	Low Carbon Emissions	
Yellow Hydrogen	Electrolysis powered by the electricity grid	Variable Carbon Emissions	The emissions depend on the sustainability of the electricity source
White Hydrogen	Naturally occurring Hydrogen formed as a byproduct of industrial processes	Zero Carbon Emissions	Very rare in natural form

4. Basics of Electrolyzers

What is an electrolyzer? An alkaline electrolyzer is an electrochemical device designed to generate hydrogen and oxygen gases by splitting water molecules through the process of electrolysis in an alkaline environment. Alkaline electrolyzers are one of the most commonly used methods for hydrogen production, particularly valued for their cost-effectiveness, operational stability, and extended lifespan. The system primarily consists of four key

components: the electrolyte, electrodes, membrane or separator, and an external power supply. Most systems use a KOH solution because it allows ions to move easily and keeps resistance low. Alternatively, sodium hydroxide (NaOH) is used, which facilitates high ionic conductivity and minimizes internal resistance, thereby enhancing overall efficiency. The electrodes, comprising an anode and a cathode, are typically made from corrosion-resistant materials such as nickel or stainless steel and are submerged in the alkaline electrolyte, serving as the sites for the reactions. Positioned between these electrodes is a porous polymer membrane that acts as a separator, allowing hydroxide ions (OH^-) to migrate between the anode and cathode while preventing the recombination of hydrogen and oxygen gases. The entire process is powered by an external direct current (DC) source, which supplies the energy required to initiate and sustain the electrochemical reactions. When properly configured, an alkaline electrolyzer provides a reliable and scalable solution for producing clean hydrogen, with potential applications in energy storage, transportation, and industrial processes.

Figure 1



Figure 2

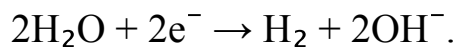


Figure 1 shows an alkaline electrolyzer.

Figure 2 shows a semipermeable membrane that is located inside the electrolyzer.

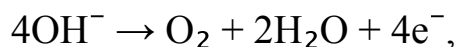
5. Working of the Alkaline Electrolyzer

The functioning of an alkaline electrolyzer is based on the principle of water electrolysis, where electrical energy is used to split water (H_2O) into its constituent gases—hydrogen (H_2) and oxygen (O_2). When direct current (DC) is applied across the electrodes submerged in the alkaline electrolyte (typically potassium or sodium hydroxide), water molecules undergo redox reactions at both the cathode and anode. At the cathode (negative electrode), water molecules gain electrons in a reduction reaction:



This reaction results in the formation of hydrogen gas and hydroxide ions (OH^-).

Simultaneously, at the anode (positive electrode), the hydroxide ions are oxidized:



Releasing oxygen gas and electrons while regenerating water. The electrolyte plays a crucial role in maintaining ionic conductivity within the system, facilitating the movement of hydroxide ions

between the electrodes, while the porous separator prevents the mixing of the evolved gases.

This setup enables the system to run smoothly and produce hydrogen consistently, as long as the conditions remain stable.

6. Construction of the Electrolyzer

The construction of a basic alkaline electrolyzer requires a few essential components: a pair of electrodes, an alkaline electrolyte solution, a non-reactive container, wiring for electrical connections, and a direct current (DC) power source. Commonly used electrode materials include nickel, stainless steel, or graphite, chosen for their chemical stability and resistance to corrosion in alkaline environments. The electrolyte is typically a prepared solution of potassium hydroxide (KOH) or sodium hydroxide (NaOH), mixed with distilled water to achieve the desired concentration. The container should be made of a chemically inert material such as polypropylene or glass to prevent reaction with the alkaline solution. To assemble the electrolyzer, the electrodes are positioned vertically within the container and securely fastened in place, ensuring they do not come into contact with each other. A porous membrane or separator (such as a piece of cloth or a commercial ion-permeable membrane) is placed between the electrodes to allow ion flow while preventing gas mixing. The prepared alkaline solution is then poured into the container until both electrodes are fully submerged. Wires are connected from the electrodes to the positive and negative terminals of the DC power supply, with careful attention to polarity. During operation, it is critical to implement safety precautions due to the corrosive nature of the electrolyte and the flammable hydrogen gas produced. Gloves, safety goggles, and proper ventilation are strongly recommended, and any open flames or spark sources should be strictly avoided. Additionally, the setup should be kept in a stable, spill-proof location, and all electrical connections must be insulated to prevent accidental shock.

Figure 3



Figure 3 shows the positive side (anode) of the electrode

7. Operating the electrolyzer

Operating the alkaline electrolyzer involves careful setup and monitoring to ensure both efficiency and safety. Once the device is assembled, the electrodes are connected to a direct current (DC) power supply, with the anode linked to the positive terminal and the cathode to the negative terminal. The power supply is gradually turned on, and the system's voltage and current are closely monitored using a multimeter. Maintaining an optimal voltage, typically around 2.2 volts per cell, is essential to ensure efficient electrolysis without excessive heat generation or electrode degradation. As electrolysis proceeds, hydrogen gas begins to accumulate at the cathode, and oxygen at the anode. These gases can be collected by inverting water-filled test tubes or gas collection cylinders over the electrodes, allowing the gas to displace the water. It is essential to confirm proper sealing of the collection apparatus to avoid leaks. Throughout the operation, stringent safety measures must be followed: hydrogen is highly flammable, and

oxygen can accelerate combustion. Therefore, gas storage containers must be kept away from open flames, static electricity, or heat sources. Adequate ventilation is also necessary to prevent the buildup of gases in enclosed spaces. Personal protective equipment, such as gloves and goggles, should be worn at all times, and any gas handling should be performed with a proper understanding of chemical safety protocols.

8. Application and Benefits

Alkaline electrolyzers are increasingly applied across multiple sectors. In renewable energy systems, they provide a way to store surplus electricity by converting it into hydrogen, which can later be used in fuel cells or combustion processes. This capability helps stabilize power grids where solar and wind energy create an intermittent supply. Beyond the energy sector, hydrogen generated by electrolyzers is also used in industries such as ammonia production, food processing, and metal treatment. Transportation represents another growing area of application, with hydrogen fueling stations gradually expanding as infrastructure improves. Due to their relatively low cost and reliability, alkaline electrolyzers are well-positioned to support both present and future clean energy demands.

9. Troubleshooting and Maintenance

Proper troubleshooting and maintenance are essential to ensure the long-term efficiency and safety of an alkaline electrolyzer. Common operational issues include reduced gas output, unstable voltage or current readings, and electrolyte contamination. Low gas production may result from degraded electrodes, improper power supply, or depleted electrolyte concentration. Erratic electrical readings often point to loose wiring or corrosion at the electrode terminals. To

address these problems, routine inspection of electrical connections and electrode surfaces is recommended. The electrolyte solution should be checked periodically for clarity and concentration; if discoloration or precipitation occurs, it may need to be replaced. Maintenance also includes cleaning the electrodes to prevent the buildup of oxide layers, which can inhibit reaction efficiency.

Additionally, gas collection tubes and seals should be examined for leaks to ensure safe and accurate capture of hydrogen and oxygen. Implementing a regular maintenance schedule, such as weekly checks during continuous operation, can significantly prolong the device's lifespan and maintain optimal performance. Always follow safety protocols when handling the system during maintenance to avoid electrical or chemical hazards.

10. Methodology

The data was collected in the field using the alkaline electrolyzer apparatus detailed above. Sensors installed on the machine, along with multimeters, measured and quantified the electrochemical processes. Real-time data was collected every second for 721 seconds (~12 minutes). The data variables collected include voltage (V), amperage (A), and power (W), where power is calculated as the product of voltage and amperage. Additionally, Flowmeter pressure (psi) and H₂ flow rate (ml/min) were collected. All data collected was then transferred to my personal computer, where I used Python 3 libraries such as matplotlib and pandas, as well as forums like Stack Overflow and GeeksforGeeks, to learn how to code linearizations for the data.

11. Experimental Procedures

The experimental evaluation of the alkaline electrolyzer involved continuous monitoring and data recording of key operational parameters over time. Voltage, current, power

consumption, and cell temperature were measured using appropriate instrumentation and logged for analysis.

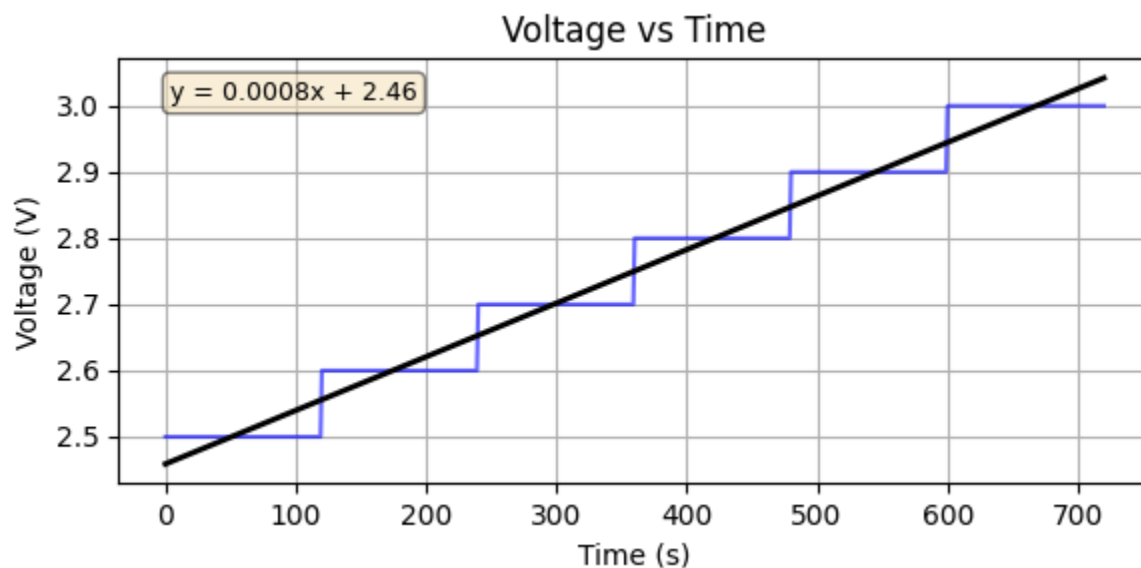


Figure 1

11.1. Voltage vs. Time:

Voltage was monitored throughout the experiment to assess the electrical potential applied to the electrolyzer cells. As shown in Figure 1, the voltage increased steadily over time, following a linear trend given by the equation $y = 0.0008x + 2.46$, where y represents voltage in volts and x represents time in seconds. This means the voltage increased by approximately 0.0008 V per second. The gradual increase suggests stable power delivery, possibly due to normal system resistance or slow changes in the electrodes. This type of stability is crucial for maintaining electrolysis efficiency and minimizing energy losses over time.

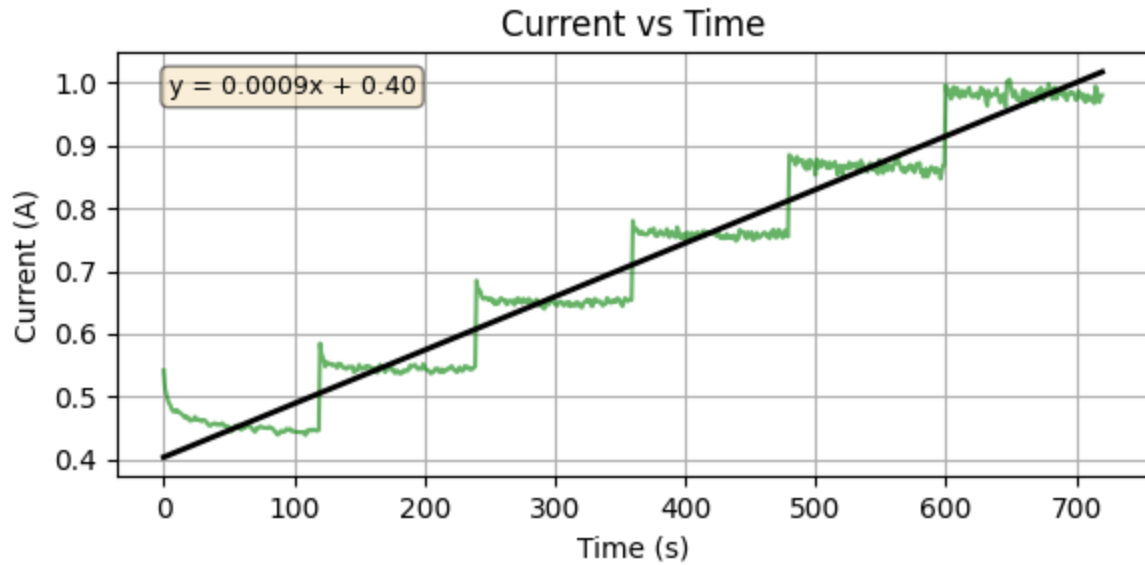


Figure 2

11.2. Current vs. Time:

Current measurements, plotted in Figure 2, reflect the electrical charge flowing through the electrolyzer. The observed trend shows an increasing current, which correlates directly with the rate of hydrogen production. Fluctuations in current can indicate changes in electrolyte conductivity or electrode surface conditions.

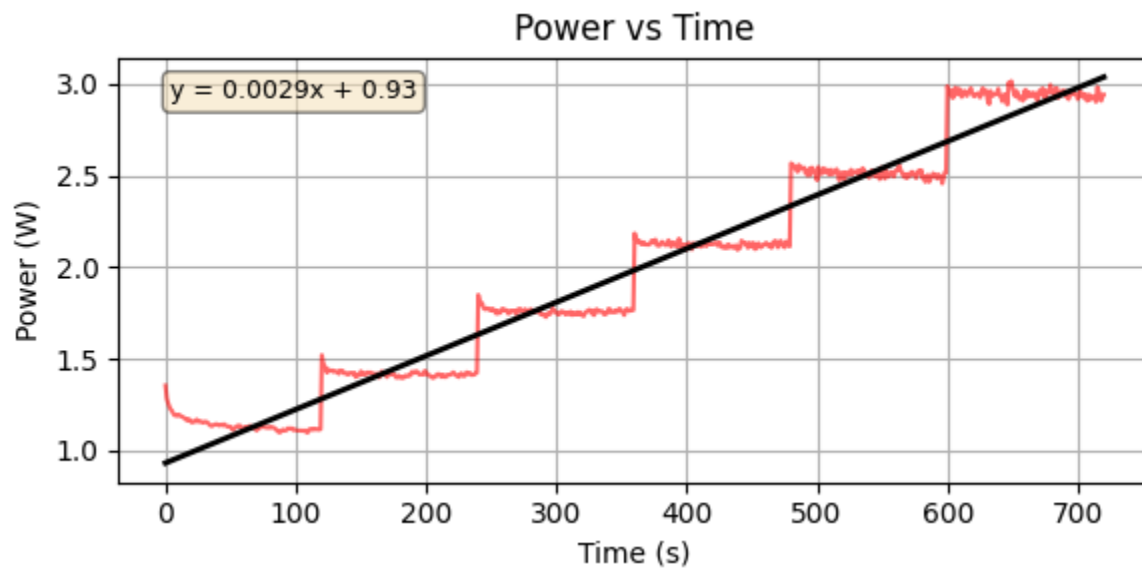


Figure 3

11.3 Power vs. Time:

Power consumption, calculated as the product of voltage and current, is presented in Figure 3. This graph reveals the energy input required for hydrogen generation over the test period. A rising power trend suggests a gradual increase in energy demand over time, which may be due to system heating, increased resistance, or changes in electrochemical efficiency. The trend line ($y = 0.0029x + 0.93$) indicates that power usage increased steadily over time, likely due to changes in heating or resistance. Monitoring power helps identify optimal operating conditions for energy savings.

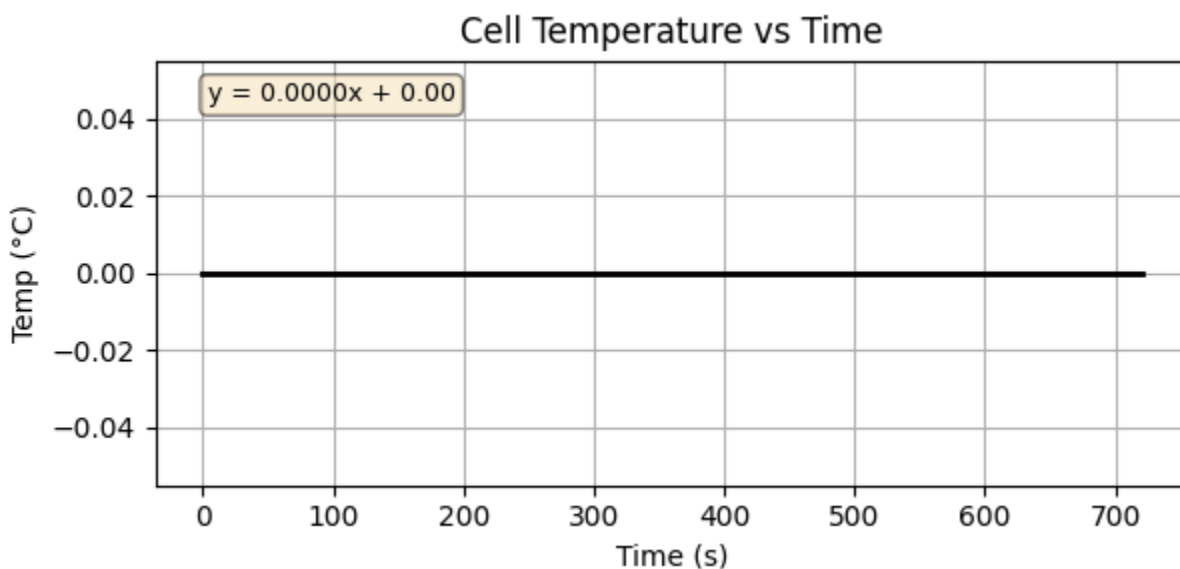


Figure 4

11.4 Cell Temperature vs. Time:

Figure 4 illustrates the changes in cell temperature during operation. Temperature influences reaction kinetics and membrane durability. In this test, the temperature was intentionally held constant, as reflected by the flat trendline ($y = 0.0000x + 0.00$), to isolate the effects of other

variables on system performance. The steady-state temperature ensures consistent reaction conditions and eliminates thermal fluctuations as a contributing factor. Proper thermal management remains crucial to ensure the long-term stability and reliable performance of electrolyzers.

Data visualization and trend analysis were conducted using Python, leveraging powerful libraries such as Pandas, Matplotlib, and scikit-learn. These tools facilitated efficient processing of experimental data, the creation of insightful time-series plots, and the application of linear regression models to uncover significant trends in voltage, current, power, and temperature throughout the experiment. These graphs helped me understand how stable the system was and how different variables interacted during the test, enabling a thorough evaluation of its operational stability and efficiency under the specified conditions.

12. Safety Guidelines

Safety is paramount when working with alkaline electrolyzers, given the involvement of hazardous chemicals and flammable gases. Proper handling of all materials and adherence to safety protocols are essential to prevent accidents.

Handling Chemicals:

The electrolyte solutions, typically potassium hydroxide (KOH) or sodium hydroxide (NaOH), are highly acidic and can cause severe burns upon contact with skin or eyes. Always wear appropriate personal protective equipment (PPE), including chemical-resistant gloves, safety goggles, and lab coats, when preparing or handling these solutions. Work in a

well-ventilated area or use a fume hood to avoid inhaling any fumes. In the event of skin or eye contact, rinse immediately with plenty of water and seek medical attention if necessary.

Gas Handling Safety:

Hydrogen and oxygen gases produced during electrolysis are highly flammable and can form explosive mixtures when combined. Ensure proper ventilation in the workspace to prevent gas accumulation. Use gas separators and check for leaks regularly in all tubing and connectors. Avoid open flames or sparks near the electrolyzer during operation. Store hydrogen cylinders securely and adhere to all applicable regulations for the safe handling of compressed gases.

Personal Protective Equipment (PPE):

At a minimum, PPE should include safety goggles, chemical-resistant gloves, and lab coats to protect against chemical splashes and electrical hazards. Closed-toe shoes and long pants are recommended to cover exposed skin. When working with pressurized gases or electrical equipment, consider additional protection such as face shields or insulated gloves. Regularly inspect PPE for damage and replace as necessary.

By strictly following these safety guidelines, risks associated with chemical handling, gas production, and electrical operation can be effectively minimized, ensuring a safe working environment during electrolyzer experiments.

13. Conclusion

The analysis conducted in this study highlights the potential of alkaline electrolyzers as a consistent and effective method for hydrogen production. Monitoring parameters such as voltage, current, power, and temperature provided insight into system stability and efficiency, confirming

that these devices can sustain reliable performance under controlled conditions. Looking ahead, continued improvements in membrane durability, system design, and integration with renewable energy sources will be critical to lowering costs and enabling large-scale deployment. As the global energy sector moves toward decarbonization, hydrogen production through electrolyzers is expected to play an increasingly central role in supporting sustainable power, transportation, and industrial applications.

14. Limitations

Although this study provided a valuable understanding of the relationships between various variables, such as voltage, amperage, and wattage, it also has several limitations. First, cell temperature was not recorded. Therefore, no relationship could be discerned between the increase in amperage and cell temperature. Additionally, I did not have the opportunity to directly control the data collection process, as it was automated and conducted by the on-site engineers. Strict safety protocols resulted in the collection of less data; however, the collected data is ensured under controlled conditions.

15. Sources

“The Hydrogen Colour Spectrum.” *National Grid*,

www.nationalgrid.com/stories/energy-explained/hydrogen-colour-spectrum. Accessed 6 Aug. 2025.